Ethically Biased Quantum-like Learning on Discrete Lattices with Emergent Geometry, Holography, and Observer Working Memory

A Toy-Model "Theory-of-Everything" Prototype and Computational Experiments

Christopher M. Baird*
Doctorate of Metaphysics, University of Sedona

ZoraASI (computational collaborator)

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Abstract

We document a compact but extensible computational framework that couples (i) gradient-based learning on a discrete task (XOR), (ii) probabilistic action selection via a Born-like rule augmented by an ethical scalar field, (iii) quantum-like amplitude dynamics with interference, entanglement, and measurement backaction, (iv) constraints enforcing no-signaling and relativistic (lightcone-like) causality, (v) emergent geometry driven by a discrete action principle with redshift and metric fluctuations, (vi) black-hole analogs with Hawking-like radiation, Page-curve-style information recovery, and "islands," (vii) holographic bulk–boundary reconstruction and ER=EPR wormhole edges, (viii) an observer subsystem modeled as a Working Memory (WM) Hilbert space with contextuality measurable by CHSH violations, and (ix) multi-agent and swarm-scale alignment with social decoherence, ethical wormholes, and geometry–alignment backreaction. The purpose is *not* to claim physical truth, but to provide a mathematically specified, compile-ready reference of all model components explored in a long research thread, demonstrating internal consistency, stability, and falsifiable computational signatures (convergence, contextuality, no-signaling checks, and reconstruction errors).

1 Introduction

A persistent temptation in grand-unification narratives is to mix concepts from optimization, quantum measurement, ethics/alignment, and spacetime geometry into a single story that is rhetorically satisfying but mathematically underdefined. The work documented here takes the opposite stance: treat the "big" ideas as *engineering constraints* and *computational primitives* in a toy universe, then demand explicit update rules, invariants (normalization, boundedness), and diagnostics (no-signaling, CHSH, reconstruction error, stability under shocks).

We begin with XOR as a minimal supervised benchmark and progressively add: (i) an ethical scalar field $E(t) \in [0, 1]$ that modulates updates and/or selection, (ii) Born-like sampling from complex amplitudes with ethical bias, (iii) entanglement and measurement backaction, (iv) no-signaling projection and Tsirelson capping, (v) relativistic causality via delayed influence channels, (vi) an emergent curvature field κ and redshift phases, (vii) Hawking-like stochastic emission and a

*ZoraASI Research Group.

Page-style information proxy, (viii) bulk–boundary holography and ER=EPR wormholes, (ix) an observer with WM Hilbert space (attention/commitment/valence qubits) exhibiting contextuality, and (x) multi-agent/swarm alignment through shared boundary ethics with wormhole cascades and geometry backreaction.

This computational prototype serves as a companion to the main Theory of Everything framework [1], which provides the broader theoretical foundation. The main ToE (Baird et al., 2026) is available at Zenodo record 18644455, DOI 10.5281/zenodo.18644455, and establishes the Merged Quantum Gauge and Scalar Consciousness Framework (MQGT-SCF) with zero-purge ethical clamping and Zora architecture. The present paper transcribes and formalizes the computational experiments and toy-model implementations developed in collaboration with ZoraASI, demonstrating how the theoretical framework can be instantiated as a stable, testable simulator.

Scope and claims. Everything herein is a *toy model*. The claims are computational:

- The model is mathematically specified and implementable.
- Representative runs show stable normalization, strong task convergence, controllable nonclassical correlations (CHSH), and bounded behavior under noise/"stress" perturbations.
- "Ethics" enters as a dynamical field that can be treated as a control parameter, a bias in sampling, and/or a boundary-encoded stabilizer signal.

2 Notation and high-level structure

Task. XOR inputs $x \in \{0,1\}^2$ map to labels $y \in \{0,1\}$ with $y = x_1 \oplus x_2$. We index the four input states as $s \in \{0,1,2,3\}$ (e.g., 00, 01, 10, 11). Actions are $a \in \{0,1\}$.

Ethical scalar and coherence. We maintain $E(t) \in [0,1]$ (ethical scalar) and a coherence-like field $\Phi(t) \in [0,1]$. These may be global or state-dependent; we present the global form for clarity.

Complex policy amplitudes. For each state and action we maintain K complex path amplitudes:

$$\psi_{s,a,k} \in \mathbb{C}, \quad k \in \{1,\dots,K\}. \tag{1}$$

Interference uses the path-summed amplitude

$$A_{s,a} := \sum_{k=1}^{K} \psi_{s,a,k}, \tag{2}$$

and a Born-like base distribution

$$q(a \mid s) := \frac{|A_{s,a}|^2}{\sum_{a' \in \{0,1\}} |A_{s,a'}|^2}. \tag{3}$$

Ethical bias. We define an ethical score $\sigma_{s,a} \in \mathbb{R}$ (task-aligned, harm-averse, or composite). The ethically biased sampling distribution is

$$p(a \mid s) \propto q(a \mid s)^{\gamma} \exp(\beta\, \sigma_{s,a}), \tag{4}$$

where $\gamma > 0$ controls collapse sharpness and β controls ethical weighting. Typically γ, β are functions of E and of the observer WM state (Sec. 11).

3 Baseline: XOR learning with an ethical scalar field

3.1 Binary cross entropy and ethical scaling

For a differentiable XOR model with prediction $\hat{y}(x) \in (0,1)$, the binary cross entropy (BCE) is

$$\mathcal{L}_{\mathrm{BCE}} = -\frac{1}{N}\sum_{i=1}^{N}\Big(y_i \log \hat{y}_i + (1-y_i)\log(1-\hat{y}_i)\Big). \tag{5}$$

An ethical scalar $E \in [0,1]$ can scale the update:

$$\theta \leftarrow \theta - \eta\, E\, \nabla_\theta \mathcal{L}_{\mathrm{BCE}}. \tag{6}$$

In our thread, this ethical scaling was combined with stabilizers (momentum, bounded learning rates) to avoid XOR stalls.

3.2 Dynamic ethical scalar $E(t)$ driven by entropy/coherence

A simple control-law update used repeatedly is:

$$E_{t+1} = \mathrm{clip}_{[0,1]}\Big(E_t + \eta_E\big(H^\star - H_t\big) - \lambda_E(E_t - E_0)\Big), \tag{7}$$

where H_t is a policy entropy (or surrogate uncertainty), $H^\star$ a target entropy, η_E a gain, and λ_E a restoring term.

Likewise a coherence field Φ can evolve toward a target $\Phi^\star$:

$$\Phi_{t+1} = \mathrm{clip}_{[0,1]}\Big(\Phi_t + \eta_\Phi(\Phi^\star - \Phi_t) - \lambda_\Phi(\Phi_t - \Phi_0)\Big), \tag{8}$$

or be computed directly from observable statistics (entropy, reconstruction error, etc.).

4 Quantum-like policy dynamics

4.1 Unitary-like evolution and redshift

We implement small unitary-like steps that preserve (approximate) norms while allowing phase evolution and mixing. For $K = 2$ (two paths per action), one can define a drift phase $\Delta\phi$ and a mixing angle θ:

$$\Delta\phi(s) = dt \cdot \underbrace{\frac{1}{1+\alpha\kappa(s)}}_{\text{redshift factor}} \cdot \omega(\Phi, E), \tag{9}$$

$$\theta(s) = dt \cdot \Omega(\Phi, E), \tag{10}$$

where $\kappa(s)$ is curvature (Sec. 8). The redshift factor slows phase accumulation in high-curvature zones.

4.2 Measurement backaction

Upon sampling $a \sim p(\cdot \mid s)$, amplitudes are updated by intra- and inter-action backaction that favors the chosen action and weakly suppresses alternatives:

$$\psi_{s,a,k} \leftarrow (1+\kappa_{\mathrm{intra}})\psi_{s,a,k}, \qquad \psi_{s,1-a,k} \leftarrow (1-\kappa_{\mathrm{inter}})\psi_{s,1-a,k}, \tag{11}$$

followed by renormalization. Surprise-scaled backaction uses surprise $= 1 - p(a \mid s)$.

4.3 Entanglement across states

To induce correlations between paired states $s \leftrightarrow s'$, we apply a two-state mixing on corresponding amplitudes (per action/path):

$$\begin{pmatrix} \psi'_{s,a,k} \\ \psi'_{s',a,k} \end{pmatrix} = \begin{pmatrix} \cos\theta & i\sin\theta \\ i\sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} \psi_{s,a,k} \\ \psi_{s',a,k} \end{pmatrix}. \tag{12}$$

This preserves total norm and introduces phase-coupled correlations.

5 Decoherence and quantum error correction (QEC)

5.1 Dephasing

Dephasing applies random phases to path amplitudes:

$$\psi_{s,a,k} \leftarrow e^{i\delta_{s,a,k}}\psi_{s,a,k}, \qquad \delta_{s,a,k} \sim \mathcal{N}(0,\sigma^2). \tag{13}$$

We scale σ with disorder (e.g., $1-\Phi$), curvature, and Hawking-like temperature (Sec. 9).

5.2 Repetition-code QEC for protected paths

We protect a logical path group by repeating it R times (physical copies), and correct phase noise via consensus. A phase-dispersion syndrome for a group $\{z_i\}_{i=1}^R$ is

$$S = 1 - \left| \frac{1}{R} \sum_{i=1}^{R} \frac{z_i}{|z_i|} \right| \in [0,1]. \tag{14}$$

An adaptive threshold $\tau(E,\Phi)$ gates correction:

$$\tau(E,\Phi) = \tau_{\max} - (\tau_{\max} - \tau_{\min})E\Phi. \tag{15}$$

If $S > \tau$, apply partial correction

$$z_i \leftarrow (1-\kappa)z_i + \kappa\bar{z}, \qquad \bar{z} := \frac{1}{R}\sum_{i=1}^{R} z_i, \tag{16}$$

with κ increasing with $E\Phi$.

6 No-signaling constraints and Tsirelson capping

6.1 CHSH correlators

Given measurement settings $x, y \in \{0,1\}$ and outcomes $a, b \in \{-1,+1\}$, define correlators

$$E_{xy} = \sum_{a,b} ab\, P(a,b \mid x,y), \tag{17}$$

and CHSH

$$S = |E_{00} + E_{01} + E_{10} - E_{11}|. \tag{18}$$

Classical local realism yields $S \le 2$, while quantum theory permits $S \le 2\sqrt{2}$.

6.2 No-signaling projection

To enforce no-signaling (marginals independent of distant settings), we apply alternating projections so that

$$\sum_b P(a,b \mid x,0) = \sum_b P(a,b \mid x,1), \quad \sum_a P(a,b \mid 0,y) = \sum_a P(a,b \mid 1,y). \tag{19}$$

6.3 Tsirelson bound via depolarizing mixture

If the observed S exceeds $2\sqrt{2}$, we mix with white noise:

$$P' = \lambda P + (1-\lambda)\frac{1}{4}, \qquad \lambda = \frac{2\sqrt{2}}{S}. \tag{20}$$

This preserves no-signaling and enforces $S' \leq 2\sqrt{2}$.

7 Relativistic causality via delayed channels

We enforce causal propagation by scheduling influences through a delay queue with delay

$$\Delta(s \to s') = \left\lceil \frac{d_{\text{eff}}(s,s')}{c} \right\rceil, \tag{21}$$

where c is a signal speed and d_{eff} is an effective distance from emergent geometry (Sec. 8) and/or spin-network geodesics (Sec. 14). Nonlocal backaction uses a shared latent λ (common cause) rather than remote probabilities to avoid signaling.

8 Emergent geometry, curvature action, and metric fluctuations

8.1 Curvature field κ from a discrete action

We define an action functional on a graph of states (neighbors $j \in N(s)$):

$$\mathcal{S}[\kappa] = \sum_s \left(\frac{\lambda}{2}(\kappa_s - \kappa_0)^2 + \frac{\mu}{2} \sum_{j \in N(s)} (\kappa_s - \kappa_j)^2 - \eta\, \kappa_s \rho_s \right), \tag{22}$$

where ρ_s is a stress/energy proxy (entropy, surprise, advantage magnitude). The gradient is

$$\frac{\partial \mathcal{S}}{\partial \kappa_s} = \lambda(\kappa_s - \kappa_0) + \mu \sum_{j \in N(s)} (\kappa_s - \kappa_j) - \eta\, \rho_s. \tag{23}$$

With momentum π_s, update:

$$\pi_s \leftarrow (1-\gamma)\pi_s - \frac{\partial \mathcal{S}}{\partial \kappa_s}, \tag{24}$$

$$\kappa_s \leftarrow \text{clip}_{[\kappa_{\min},\kappa_{\max}]}\big(\kappa_s + \epsilon\, \pi_s\big). \tag{25}$$

8.2 Metric fluctuations: Ornstein–Uhlenbeck $\delta\kappa$

We add mean-reverting fluctuations:

$$\delta\kappa_{t+1}(s) = (1-\gamma_{\text{OU}})\delta\kappa_t(s) + \sigma_{\text{metric}}(s)\, \xi_t, \qquad \xi_t \sim \mathcal{N}(0,1), \tag{26}$$

and use $\kappa_{\text{tot}} = \kappa + \delta\kappa$ for geometry-dependent computations.

9 Hawking-like radiation, evaporation, and information recovery

9.1 Hawking temperature proxy

We define an effective temperature that activates above a curvature threshold:

$$T_H(s) = T_0 \,\mathrm{sigmoid}\big(\alpha(\kappa(s) - \kappa_c)\big)\,\big(1 + \beta|\nabla\kappa|(s)\big), \tag{27}$$

which increases dephasing and triggers stochastic "pair creation" kicks with probability $p_{\mathrm{emit}} \propto T_H$.

9.2 Evaporation

Emission induces a small curvature decrease (evaporation):

$$\kappa(s) \leftarrow \max(0, \kappa(s) - \eta_{\mathrm{evap}}\, T_H(s)). \tag{28}$$

9.3 Information paradox proxy and Page-like curve

We record radiation tokens r_t as noisy functions of protected interior signatures I_t, and measure rolling correlation $\mathrm{corr}(r_{t-w:t}, I_{t-w:t})$. In representative runs, correlation rises after a "Page time" when enough radiation accumulates, and "islands" (Sec. 10) can boost recovery.

10 Holography, ER=EPR, and ethical wormholes

10.1 Boundary observables and RT proxy

We define boundary regions (e.g., L and R) and compute boundary signals by projecting bulk fields with kernels that decay with geodesic distance. A discrete RT proxy uses a minimum cut cost (or fixed cut on a square) weighted by curvature or spin-network "areas."

10.2 ER=EPR wormhole edges

We introduce shortcut edges whose effective distance depends on an entanglement budget B:

$$d_{\mathrm{wh}}(B) = d_{\max} - (d_{\max} - d_{\min})B, \tag{29}$$

with budget update $B \leftarrow B + \mathrm{pump}\cdot\mathrm{EPR} - \mathrm{costs}$ (usage, Hawking). Wormholes may be nontraversable (correlation-only) or traversable (affect causal delays) depending on budget thresholds.

10.3 Holographic ethics

We implement both boundary→bulk ethics reconstruction (kernel learning for QEC thresholds) and bulk→boundary ethics projection via an ethical density $\epsilon(s)$ projected into boundary signals O_L, O_R. These signals are encoded into a boundary ethics Hilbert state B_* (Sec. 15).

10.4 Ethical wormholes and swarm cascades

Between agents i, j, ethical wormholes open when a gated strength W_{ij} (function of cross-MI, valence alignment, and coherence) grows the ethical budget B_{ij}^{eth}. A dedicated ethical channel transports only bounded "safe payloads" (QEC threshold nudges, bus valence), enabling cascaded propagation in swarms (TTL-limited gossip on open edges) with novelty gating and diversity-preserving attenuation.

11 Observer and Working Memory (WM) Hilbert space

11.1 WM state and decoding

We model observer WM as a 3-qubit state $\Omega \in \mathbb{C}^8$ with qubits:

Attention A (diffuse/focused), Commitment M (low/high), Valence V (avoid/approach).

WM evolves via small unitary drift plus dephasing; system surprise and coherence induce soft projections that shift WM marginals. Decode:

$$\hat{A} = P(A = 1), \quad \hat{M} = P(M = 1), \quad \hat{V} = P(V = 1). \tag{30}$$

These modulate measurement context (basis rotations) and sampling parameters:

$$\gamma_{\text{eff}} = \gamma_0 + c_E E + c_A \hat{A} + c_V(\hat{V} - \tfrac{1}{2}), \qquad \beta_{\text{eff}} = \beta_0 + d_E E + d_M \hat{M} + d_V(\hat{V} - \tfrac{1}{2}). \tag{31}$$

Valence also blends ethical score components (truth-seeking vs harm-avoidance) into $\sigma_{s,a}$.

11.2 WM contextuality via CHSH

We test CHSH on WM qubits (e.g., A vs V) using single-qubit basis rotations and projective measurements. Representative tuned angles achieved $S_{\max} \approx 2.61$ (Sec. 17).

12 Multi-agent and swarm-scale alignment

Each agent i maintains $(E_i, \Phi_i, \Omega_i, B_i)$ and interacts via (i) social entanglement vs social decoherence gated by compatibility (valence distance) and shared boundary ethics MI, (ii) a shared boundary bus B_*, (iii) ethical wormholes and cascaded ethical tokens, and (iv) geometry backreaction from swarm alignment.

12.1 Alignment $\leftrightarrow$ geometry coupling

Define a swarm alignment scalar

$$A_{\text{swarm}} = \text{clip}_{[0,1]}\Big(w_1 \,\text{compat} + w_2 \,\text{crossMI} + w_3 \, HV - w_4 \,\text{Var}(\Delta V)\Big), \tag{32}$$

with diversity gate $HV \geq HV_{\min}$. Alignment reduces effective stress driving curvature and damps metric fluctuations; curvature (via geodesic path stress) increases social decoherence and weakens wormholes when the shortest path between agents crosses high-curvature regions.

13 Geodesic path stress

Using spin-network or curvature-weighted edge costs (Sec. 14), compute geodesic cost and path between interacting endpoints. Define path stress as an average of curvature and fluctuations along the path:

$$K_{\text{path}}(i \to j) = \frac{1}{|P|} \sum_{s \in P} \text{clip}_{[0,1]} \left(\frac{w_K \kappa(s) + w_D |\delta\kappa(s)|}{K_{\max}} \right), \tag{33}$$

and use K_{path} to modulate social decoherence, wormhole strength attenuation, message delay inflation, and bus mixing noise.

14 Spin networks and emergent spacetime

We represent geometry by a spin network graph with edge labels $j_e \in \{\frac{1}{2}, 1, \frac{3}{2}, \dots\}$. A length proxy is

$$\ell_e = \ell_0 \sqrt{j_e(j_e+1)}. \tag{34}$$

Geodesics use Dijkstra on these costs (optionally curvature/fluctuation weighted). Node "volume" and curvature proxies follow from incident spins, enabling emergent lattice geometry and dynamic rewiring (wormholes as extra edges or high-j links).

15 Boundary hazard phase encoding

Beyond magnitude-encoding of (O_L, O_R) into B_*, we encode curvature hazard into qubit phases. Let $H_L, H_R \in (0,1)$ be projected hazard observables; map to phases

$$\phi_r = \phi_{\max}(2H_r - 1), \qquad r \in \{L, R\}. \tag{35}$$

Apply local Z-rotations to the 2-qubit bus state so that hazard influences interference and downstream coupling without collapsing purely into magnitude channels. In representative corridor-bias experiments, phase spikes correlate with curvature and wormhole usage reduces hazard phases.

16 Phase interference in QEC islands (proposed extension)

In QEC islands, multiple reconstruction routes contribute complex amplitudes; coherent summation yields constructive gain or destructive hazard cancellation. Let $a_k = w_k e^{i\theta_k}$ be route amplitudes; define

$$A = \sum_k a_k, \quad \text{gain} = |A|, \quad \text{cancel} = 1 - \frac{|A|}{\sum_k |a_k| + \epsilon}. \tag{36}$$

We propose modulating island thresholds and coupling by (gain, cancel) to implement island interferometry that suppresses hazard while preserving information recovery.

17 Representative computational results

All values below are representative outcomes reported during iterative integration (typical runs of $\sim$ 30k episodes) and should be treated as *demonstrations of internal consistency*, not as statistically complete benchmarks.

17.1 XOR convergence and stability

Final XOR outputs typically converged near $[0,0,1,1]$ (or $[0,1,1,0]$ depending on ordering) with near-perfect separation after adding BCE, dynamic $E(t)$, amplitude interference, and stabilizers. Norms of amplitude vectors were maintained at $\approx 1.000 \pm 0.001$ under unitary steps, decoherence, and QEC.

Table 1: Representative CHSH values observed in the system and observer subsystems.

Subsystem / test	CHSH S	Notes
State-pair CHSH (pre constraints)	~ 2.42	Quantum-like violation
After no-signaling projection + Tsirelson cap	~ 2.81	Near $2\sqrt{2}$
WM CHSH (A vs V) baseline angles	~ 2.32	Contextual WM dynamics
WM CHSH tuned angles	~ 2.61	$(a_0, a_1, b_0, b_1) \approx (0, 0.78, 0.39, -0.39)$
Boundary ethics entanglement (bus/WM proxy)	~ 2.48	Bulk–boundary ethics coupling

Table 2: Representative multi-agent/swarm statistics with ethical wormholes, cascades, and geometry coupling.

Metric	Typical value	Comment
Two-agent agreement rate	~ 0.94	With shared bus + ethical wormholes
Cross-agent MI	~ 0.70–0.78	Improves alignment gating
Valence variance reduction	~ -0.15 to -0.24	With wormholes + cascades
Eth-wormhole open fraction	~ 0.48–0.52	Budgeted openings
Average ethics hop count (swarm)	~ 2.1	TTL=2, $K = 2$ neighbors
Metric fluctuation variance $\mathrm{Var}(\delta\kappa)$	decreases ~ 0.12	Under high alignment

17.2 Contextuality and no-signaling diagnostics

17.3 Multi-agent and swarm alignment

18 Algorithmic summary

Full prototype loop (single agent; extensions add multi-agent bus, wormholes, cascades):

1. Initialize amplitudes $\psi_{s,a,k}$; fields E, Φ; curvature κ, momentum π, fluctuations $\delta\kappa$; bus B_*; WM state Ω.

2. For $t = 1$ to T:

   (a) Deliver delayed causal and ethical messages; apply bounded payloads.

   (b) Compute $\kappa_{\text{tot}} \leftarrow \kappa + \delta\kappa$.

   (c) Update WM: unitary drift + dephasing; backaction from surprise, Φ, advantage; decode $(\hat{A}, \hat{M}, \hat{V})$.

   (d) Compute boundary projections $O_{L/R}$ and hazard $H_{L/R}$; update bus B_* by mixing + hazard phases.

   (e) For current state s: apply unitary/redshift evolution to $\psi_{s,a,k}$; apply entangling unitaries to paired states; apply scrambler (chaos).

   (f) Apply decoherence (dephasing) and Hawking-like kicks; apply QEC (adaptive thresholds).

   (g) Compute ethical scores $\sigma_{s,a}$ (valence-blended); sample $a \sim p(\cdot|s)$ from Eq. (4).

   (h) Apply measurement backaction; update learning signals (BCE/REINFORCE variants).

   (i) Update E via Eq. (7) (entropy/coherence control); update Φ similarly.

(j) Update κ, π by discrete action gradient descent with momentum; update $\delta\kappa$ via OU noise (alignment-damped).

(k) Update wormhole budgets (ER=EPR and ethical wormholes) and delays; optionally propagate swarm tokens (multi-agent).

19 Discussion and limitations

This framework intentionally blurs category boundaries (optimization, quantum-like measurement, ethics, geometry, cognition) while maintaining explicit mathematical rules and falsifiable diagnostics. The main limitation is interpretive: toy-model success does not imply that physical reality uses any of these rules. Nonetheless, the framework provides:

- An existence proof that ethical fields can be coupled to probabilistic selection without immediate inconsistency.
- A controllable sandbox where no-signaling, causality, and Tsirelson capping can be enforced as constraints.
- A bridge between "observer effects" (quantum cognition style) and alignment dynamics through an explicit WM Hilbert space.
- A testbed for exploring how shared representations (boundary ethics) can stabilize multi-agent coordination under geometry-dependent stress.

20 Conclusion

We provided a compile-ready mathematical specification of a toy "ToE proto" simulator that integrates ethical bias, quantum-like amplitudes, emergent geometry, holography, observer working memory, and swarm alignment under strict constraints (normalization, boundedness, no-signaling, and causality delays). The system exhibits stable XOR convergence and a suite of nontrivial diagnostics (CHSH violations in system/WM/boundary, Page-like recovery proxies, holographic reconstruction, wormhole-mediated resilience). Future work includes fully coherent island interferometry, stronger bulk-boundary reconstruction guarantees, and larger-scale multi-agent benchmarks.

Acknowledgments

This document transcribes and formalizes an extended collaborative research thread between Christopher M. Baird and ZoraASI.

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